Monday Lab Set A

Q1 Given an integer array, find the contiguous subarray within it that has the maximum product.
Input: arr = {2, 3, -2, 4}
Output: 6 {The subarray {2, 3} has the maximum product 6.}

Q2 A new phone is priced at Rs. 20,000. Every year, the phone undergoes a sudden price drop given by the formula, price_drop=(current_price-12000) ^{2}/20000. Write a recursive function price() to calculate the price for every year. The function should take the number of years n as an argument, calculate the price after n years, and return the result. The function should be implemented as a program that reads the number of years as input, determines the price using the price() function and prints the result.

Sample input 1: 1
Sample output 1: 16800

Sample input 2: 3
Sample output 2: 14892.6

Q3 Problem Statement: Given an array where each element represents the maximum number of steps you can jump forward, find the minimum number of jumps required to reach the end of the array. Use recursion to solve the problem.

Input: Array: [2, 3, 1, 1, 2, 4, 2, 0, 1, 1]
Output: Minimum Jumps: 4

Explanation of the Output: The minimum jumps required to reach the end are 4: 2 -> 3 -> 2 -> 4 -> End.
Let's break down how to reach the end of the array (the last index) effectively:
Start at Index 0: The value is 2, meaning you can jump to either Index 1 or index 2. Jump to Index 1 (first jump): From index 1, the value is 3, so you can jump to indices 2, 3, or 4. Jump to Index 4 (second jump): From index 4, the value is 2, allowing you to jump to indices 5 or 6. Jump to Index 5 (third jump): From index 5, the value is 4, allowing you to jump to indices 6, 7, 8, or 9. Finally, Jump to Index 9 (fourth jump): Index 9 is the last index of the array, and you have successfully reached the end.
Jumps Taken: From Index 0 to Index 1 → First Jump From Index 1 to Index 4 → Second Jump From Index 4 to Index 5 → Third Jump From Index 5 to Index 9 → Fourth Jump Based on this path, the minimum number of jumps required to reach the end of the array is 4.

Monday Lab Set B

Q1 Write a recursive function float harm_rec(int n) which computes the sum of the first n terms of harmonic series (n is passed in parameter):

$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n}$

Q2 Suppose you have an array cost[ ] of size n where cost[i] is the cost of the ith step on a staircase. Once you pay the cost, you can either climb one or two steps of the stairs. You can either start from the step with index 0 or step with index 1. Write a function minCost(int cost[ ], int n) that computes and returns an integer value, which is the minimum cost incurred in reaching the top floor.

Sample Input

Enter the number of elements in the array : 3

10 15 20

Sample Output Minimum Cost = 15

Explanation: Cheapest is: start on cost[1], pay that cost, and go to the top.

Sample Input 1

10

1 100 1 1 1 100 1 1 100 1

Sample Output 1

Minimum Cost = 6

Explanation: Cheapest is: start on cost[0], and only step on 1s, skipping cost[3].

Hint: minCost[i] = cost[i] + min(minCost[i - 1], minCost[i - 2]) for i ¿= 2. min-Cost[0] = cost[0], minCost[1] = cost[1]

Q3 Implement the next permutation, which rearranges the list of numbers into Lexicographically next greater permutation of list of numbers. If such arrangement is not possible, it must be rearranged to the lowest possible order i.e. sorted in an ascending order. You are given an list of numbers arr[ ] of size N.

Example 1:

Input: N = 6

arr = {1, 2, 3, 6, 5, 4}

Output: {1, 2, 4, 3, 5, 6}

Explanation: The next permutation of the given array is {1, 2, 4, 3, 5, 6}.

Example 2:

Input: N = 3

arr = {3, 2, 1}

Output: {1, 2, 3}

Explanation: As arr[] is the last permutation. So, the next permutation is the lowest one.

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